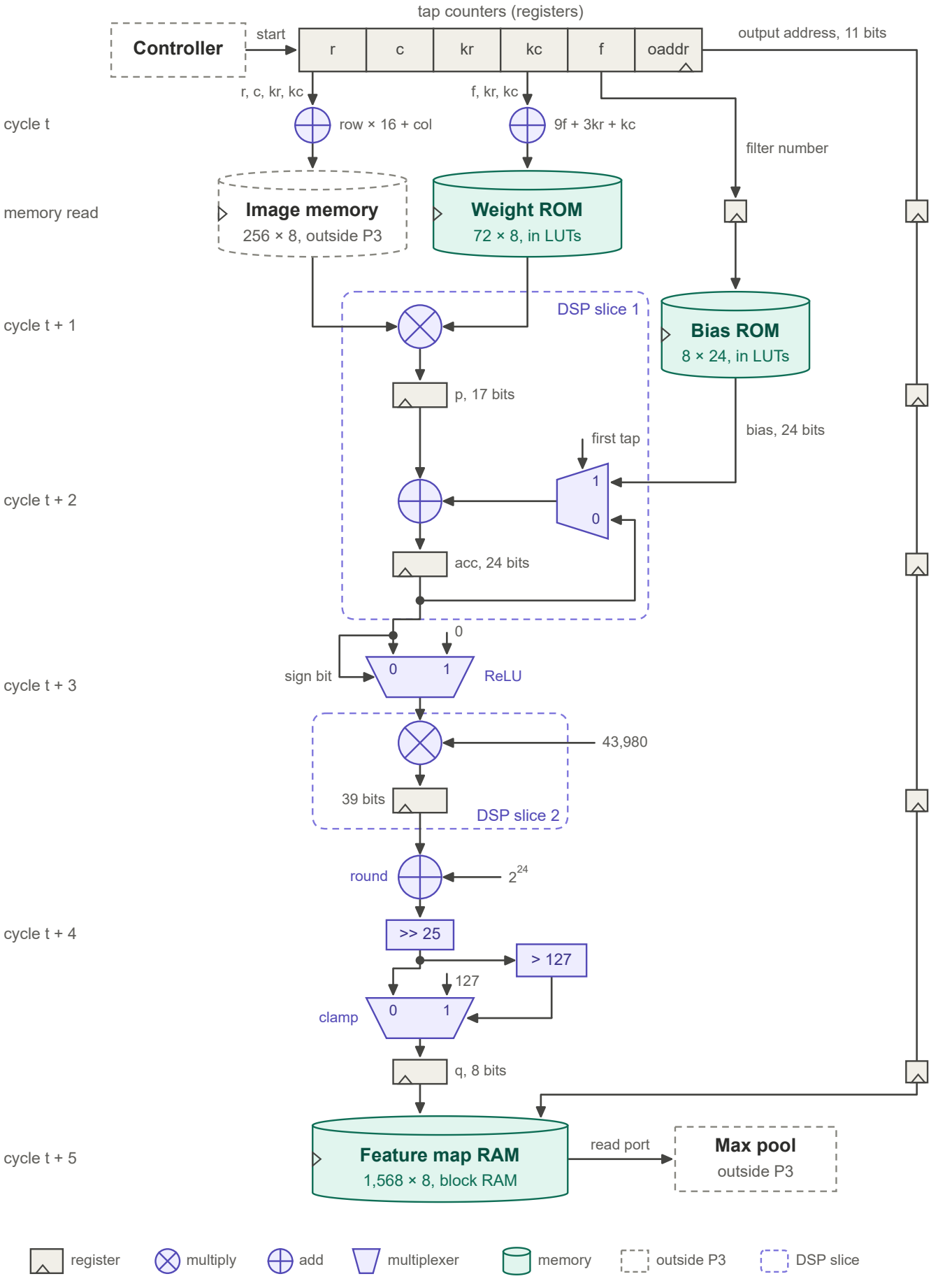

tap counters (registers)
Controller
start
r
c
kr
kc
f
oaddr
output address, 11 bits
r, c, kr, kc
f, kr, kc
cycle t
row × 16 + col
9f + 3kr + kc
filter number
memory read
Image memory
256 × 8, outside P3
Weight ROM
72 × 8, in LUTs
cycle t + 1
DSP slice 1
Bias ROM
8 × 24, in LUTs
p, 17 bits
bias, 24 bits
first tap
cycle t + 2
1
0
acc, 24 bits
0
sign bit
0
1
ReLU
cycle t + 3
43,980
39 bits
DSP slice 2
round
2^24
cycle t + 4
>> 25
> 127
127
clamp
0
1
q, 8 bits
cycle t + 5
Feature map RAM
1,568 × 8, block RAM
read port
Max pool
outside P3
register
multiply
add
multiplexer
memory
outside P3
DSP slice